Module 1: Foundations, History & Philosophical Underpinnings (Q1–Q7)

1. Define Artificial Intelligence. Differentiate between the four main approaches to defining AI: acting humanly, thinking humanly, acting rationally, and thinking rationally. Which approach is most widely adopted in modern AI research, and why?

2. Trace the historical evolution of AI from the Dartmouth Conference of 1956 to the present day. Highlight at least two major "AI winters" and the key factors that led to the current "AI spring."

3. Explain the Turing Test. What is its primary significance in evaluating machine intelligence? Discuss at least two major criticisms of using the Turing Test as a definitive measure of intelligence.

4. Differentiate between Artificial Narrow Intelligence (ANI), Artificial General Intelligence (AGI), and Artificial Superintelligence (ASI). Provide practical, real-world examples of ANI currently in use.

5. Compare and contrast the "Symbolic AI" (Good Old-Fashioned AI – GOFAI) paradigm with the "Connectionist" (neural network) paradigm. What are the inherent strengths and weaknesses of each?

6. Discuss the Chinese Room Argument proposed by John Searle. How does this thought experiment challenge the idea that a computer executing a program possesses true understanding, consciousness, or intentionality?

7. Evaluate the contributions of Alan Turing and John McCarthy. How did their distinct visions (Turing's focus on computation and McCarthy's focus on logic) shape the dual trajectories of the AI field?

Module 2: Intelligent Agents and Environments (Q8–Q12)

8. Define an "Intelligent Agent." What are the fundamental components of a rational agent, and what does "rationality" mean in the context of AI decision-making?

9. Describe the PEAS framework (Performance, Environment, Actuators, Sensors). Apply this framework to thoroughly design an intelligent agent for a fully autonomous robotic surgical system.

10. Explain the difference between a fully observable environment and a partially observable environment. Provide one detailed AI application example for each and discuss how the observability impacts the agent's internal model.

11. Distinguish between deterministic and stochastic environments in AI. How does the degree of stochasticity influence the agent's choice of decision-making algorithms (e.g., search vs. probabilistic reasoning)?

12. Contrast single-agent environments with multi-agent environments. In multi-agent settings, distinguish between cooperative and competitive scenarios. Provide an AI application for each type of interaction.

Module 3: Problem Solving, Uninformed & Informed Search (Q13–Q20)

13. What constitutes a well-defined search problem? Define all key components: state space, initial state, actions, transition model, goal test, and path cost.

14. Explain the Breadth-First Search (BFS) algorithm in detail. Under what specific conditions is BFS considered optimal, and what are its primary limitations regarding time and memory complexity in large search spaces?

15. Analyze Depth-First Search (DFS) and its iterative deepening variant. In what specific problem scenarios (by tree structure or depth constraints) would you prefer DFS over BFS, despite its incompleteness in infinite spaces?

16. Discuss the concept of an informed (heuristic) search strategy. Explain what a heuristic function is, and mathematically define the conditions for an admissible and consistent heuristic.

17. Describe the A search algorithm step-by-step. Why is A considered optimally efficient when using an admissible heuristic? Explain the significance of the "f = g + h" cost function.

18. Differentiate between hill-climbing (greedy local search) and simulated annealing. What is the primary drawback of simple hill-climbing, and how does simulated annealing mathematically attempt to mitigate this issue (e.g., using temperature and probability)?

19. Define adversarial search and its application in game-playing. Explain the Minimax algorithm. How does it guarantee optimal play for a rational agent in a zero-sum game?

20. Describe Alpha-Beta Pruning. How does this optimization technique improve the efficiency of the Minimax algorithm, and crucially, why does it not affect the final optimal decision?

Module 4: Knowledge Representation and Reasoning (Q21–Q25)

21. Define Knowledge Representation (KR) and explain why it is a critical subfield of AI. Briefly describe the trade-offs between three major KR formalisms (e.g., Logic, Semantic Nets, and Frames).

22. Explain Propositional Logic and First-Order Logic (FOL). What specific advantages does FOL provide over propositional logic in terms of representing quantifiers, objects, and relations in complex domains?

23. What is an inference engine? Explain the processes of forward chaining (data-driven) and backward chaining (goal-driven) in rule-based expert systems. Provide a scenario where backward chaining is more efficient.

24. Differentiate between deductive, inductive, and abductive reasoning in the context of AI. Provide an AI use case for each type (e.g., expert systems, machine learning generalization, and diagnostic systems).

25. How are Semantic Networks and Frames used to structure knowledge? Discuss how they handle inheritance, instance relationships, and slot defaults. What are the limitations of these semantic representations compared to formal logic?

Module 5: Foundational Machine Learning (Q26–Q33)

26. Define Machine Learning (ML) and explicitly explain how it fundamentally differs from traditional deterministic programming. Why is ML particularly suitable for perceptual and high-dimensional tasks (e.g., vision or speech)?

27. Illustrate the three main paradigms of machine learning: Supervised Learning, Unsupervised Learning, and Reinforcement Learning. Provide one complex, real-world application for each and justify why that paradigm is the best fit.

28. Explain the Bias-Variance Tradeoff in supervised learning. Mathematically and conceptually, describe what happens when a model exhibits high bias versus when it exhibits high variance.

29. Describe the concepts of overfitting and underfitting. What are the common techniques (e.g., cross-validation, regularization, pruning) used to prevent overfitting in statistical ML models?

30. Compare and contrast Classification and Regression tasks in supervised learning. Give specific examples of evaluation metrics used for each (e.g., Accuracy/F1-score vs. RMSE/MSE).

31. What is the K-Nearest Neighbors (KNN) algorithm? How does the choice of the 'K' hyperparameter and the distance metric (e.g., Euclidean vs. Manhattan) critically affect the classifier's decision boundary and performance?

32. Explain the k-means clustering algorithm. What is the "elbow method," and how is it used to determine the optimal number of clusters? Discuss the algorithm's sensitivity to initial centroid selection.

33. Define Reinforcement Learning (RL) and its core components: Agent, Environment, State, Action, Reward, and Policy. How does the fundamental goal of maximizing cumulative reward differ from immediate reward-seeking behavior?

Module 6: Neural Networks and Deep Learning (Q34–Q38)

34. What is an Artificial Neural Network? Explain the biological inspiration behind the artificial neuron and describe the mathematical operation performed within a single artificial neuron (weighted sum + activation).

35. Describe the structure of a simple Perceptron. What is the fundamental mathematical limitation of a single-layer perceptron (the XOR problem)? Explain how Minsky and Papert's proof of this limitation affected the trajectory of AI in the late 1960s.

36. Explain the structure of a Multi-Layer Perceptron (MLP). How does the inclusion of one or more hidden layers overcome the linear separability constraint of the single perceptron (Universal Approximation Theorem)?

37. What is an Activation Function? Discuss the mathematical properties and specific use-cases of three different activation functions: Sigmoid, Hyperbolic Tangent (Tanh), and Rectified Linear Unit (ReLU). What specific problem does ReLU solve in deep networks (vanishing gradient)?

38. Define Backpropagation. Provide a high-level mathematical intuition of how the chain rule is utilized to calculate gradients and update the weights in a neural network during training. What is the role of the learning rate?

Module 7: Natural Language Processing & Computer Vision (Q39–Q43)

39. What is Natural Language Processing (NLP)? List and explain at least four major challenges in processing human language for machines (e.g., lexical ambiguity, syntactic ambiguity, coreference resolution, and sarcasm/detection).

40. Describe the Bag-of-Words (BoW) model and Term Frequency-Inverse Document Frequency (TF-IDF). What are the strengths of these traditional text representation methods, and what is their primary weakness regarding semantic meaning and word order?

41. Differentiate between syntax and semantics in the context of NLP. Explain how syntactic parsing helps identify sentence structure, and how Named Entity Recognition (NER) contributes to extracting semantic information.

42. Define Computer Vision. What are the primary intrinsic challenges in enabling machines to interpret visual data from the real world (e.g., illumination variation, occlusion, viewpoint variance)?

43. Explain how Convolutional Neural Networks (CNNs) are specifically designed to process grid-like image data. Detail the distinct roles of the convolutional layer (feature extraction), the pooling layer (dimensionality reduction), and the fully connected layer (classification) in a typical CNN architecture.

Module 8: Ethics, Philosophy, Safety, and the Future (Q44–Q50)

44. Discuss the major ethical concerns associated with the widespread deployment of AI systems in society (e.g., algorithmic bias, mass surveillance, autonomous weaponry, and job displacement). How are these concerns interconnected?

45. What is algorithmic bias? Provide a detailed example of how biased training data can lead to discriminatory outcomes in a real-world AI system (e.g., in hiring, lending, or criminal recidivism scoring). What steps can mitigate this?

46. Explain the concept of "Explainable AI" (XAI). Why is interpretability critically important in high-stakes domains like healthcare diagnostics and autonomous driving? Discuss the trade-off between high accuracy (black-box models) and explainability.

47. Discuss the short-term and long-term future outlook of AI in the next decade. What are the major technological bottlenecks (e.g., data requirements, energy consumption, lack of common sense) that must be overcome to move closer to AGI?

48. What is the technological singularity hypothesis, as popularized by Ray Kurzweil? Critically analyze the feasibility of this event occurring within this century, considering both computational limits and our understanding of human consciousness.

49. Choose a specific industry (e.g., personalized medicine, autonomous agriculture, or climate change modeling). Critically evaluate how AI is currently transforming operations in this industry and discuss the primary limitations or risks associated with this transformation.

50. Synthesize your understanding: What does it mean for an AI system to be truly "robust" and "safe" in an open-world setting? Propose a multi-layered framework (covering technical, regulatory, and ethical measures) to ensure the safety and robustness of AI systems before they are deployed in critical national infrastructure.

51. State and explain any three applications of Artificial Intelligence in modern society.

52. What is Machine Learning? Briefly explain:
    a. Supervised Learning
    b. Unsupervised Learning
53. Define an Expert System. Identify and explain its two major components.
54. Explain the difference between Narrow AI and General AI with examples.
55. What is Natural Language Processing (NLP)? Mention any three real-life applications of NLP.
56. State any three advantages and three disadvantages of Artificial Intelligence.
57. Explain the role of AI in the following areas:

- Healthcare
- Education
- Banking

SPECIFIC PROBLEMS

Question 1

What is an Intelligent Agent? Mention any two characteristics of intelligent agents.

Question 2

Explain the difference between Supervised Learning and Reinforcement Learning.

Question 3

Define the term "Expert System". State any two areas where expert systems are used.

Question 4

What is Natural Language Processing (NLP)? Explain any two NLP applications.

Question 5

State any three ethical issues associated with Artificial Intelligence.

Question 6

Explain the importance of Artificial Intelligence in the following sectors:
a) Agriculture
b) Security
c) Transportation

PROBLEM SCENARIOS

Question 1: AI in Agriculture

A farming company wants to use AI systems to monitor crops, predict rainfall, and detect plant diseases automatically using drones and sensors.

Tasks

a) Explain how AI can improve agricultural productivity. (3 marks)
b) Identify two AI technologies used in this system. (2 marks)
c) State three benefits of AI in agriculture. (3 marks)
d) Mention two challenges farmers may face when implementing AI solutions. (2 marks)

Question 2: AI-Based E-Learning System

A university develops an AI-powered e-learning platform that recommends learning materials to students based on their performance and learning speed.

Tasks

a) Define personalized learning in AI. (2 marks)
b) Explain how Machine Learning can help the platform recommend materials. (3 marks)
c) State three advantages of AI in education. (3 marks)
d) Mention two limitations of AI-based learning systems. (2 marks)

Question 3: AI in Healthcare

A hospital wants to introduce an AI-based system capable of detecting diseases from patients' medical records and laboratory results. The system should help doctors diagnose illnesses faster and reduce medical errors.

Tasks:

a) Identify the type of AI system suitable for this scenario. (2 marks)
b) Explain how Machine Learning can help improve diagnosis accuracy. (3 marks)
c) State two benefits of using AI in healthcare. (2 marks)
d) Mention three possible challenges the hospital may face when implementing the AI system. (3 marks)

Question 4: AI Chatbot for a University

A university plans to deploy an AI chatbot to answer students' questions about admission, fees, results, and course registration.

Tasks:

a) Define a chatbot. (2 marks)
b) Identify the AI technology used by the chatbot. (2 marks)
c) Explain how Natural Language Processing helps the chatbot communicate with students. (3 marks)
d) State three advantages of using AI chatbots in universities. (3 marks)

Question 4: AI-Based Smart Traffic System

A city experiences heavy traffic congestion daily. The government decides to implement an AI-powered smart traffic management system that uses cameras and sensors to control traffic lights automatically.

Tasks:

a) Explain how AI can help reduce traffic congestion. (3 marks)
b) Identify two AI technologies that may be used in the system. (2 marks)
c) State three benefits of the smart traffic system. (3 marks)
d) Mention two risks or limitations of using AI in traffic management. (2 marks)
